

Physical app plan , by Erol Cemiloglu

This is a full high level concept description of my app called “Physical” to be made. It will exist as a ios/andriod app and presented alongside google health & fitbit inspire 3, which are connected via api.

Part 1: Logs

We will be taking a variety of different data ranging from food/nutrition, strength levels, performance, vitals/recovery, chassis and aesthetics.

Some will be manually logged on the app but some also are automatically connected via api of google health and fitbit inspire 3.

Table 1 references the full range of data incorporated, and how it will be taken in, feedback is open to improving this table.

Table 1:

Data metric:	Data category:	Data source:
Chest strength	strength (reps & weight) - yielding volume & 1repmax score.	via bench press manual log on the physical app (note exercise selection is based on most apt to reference on population strength levels and representative of the actual strength)
Front Shoulder strength	strength (reps & weight) - yielding volume & 1repmax score.	Via overhead press manual log on the physical app
Medial shoulder strength	strength (reps & weight) - yielding volume & 1repmax score.	Via lateral raise manual log on the physical app
Bicep strength	strength (reps & weight) - yielding volume & 1repmax score.	Via bicep curl manual log on the physical app
Tricep strength	strength (reps & weight) - yielding volume & 1repmax score.	Via tricep skull crusher manual log on the physical app.
Forearm strength	strength (reps & weight) - yielding volume & 1repmax score.	Via forearm curl manual log on the physical app.
Latissumus dorsi strength	strength (reps & weight) - yielding volume & 1repmax score.	Via weighted/assisted pullup manual log on the physical app
Glute strength	strength (reps & weight) -	Via barbell hip thrust manual log

	yielding volume & 1repmax score.	on the physical app
Quadriceps strength	strength (reps & weight) - yielding volume & 1repmax score.	Via the squat manual log on the physical app
Hamstrings strength	strength (reps & weight) - yielding volume & 1repmax score.	Via the Romanian deadlift manual log on the physical app.
Calves strength	strength (reps & weight) - yielding volume & 1repmax score.	Via the standing calf raise manual log on the physical app.
Abs strength	strength (reps & weight) - yielding volume & 1repmax score.	Via the abs crunch manual log on the physical app.
Lifting exercise sets	Strength (background)	Manually logged in physical in conjunction with confirmation of a workout occurring from the google health app. Yielding knowledge of strength progress, volume and hit muscle groups for improved context.
Sleep score	Recovery	Automatically logged daily from the api from google health and the inspire 3
Heart rate variability	Recovery	Automatically logged daily from the api from google health and the inspire 3
Resting heart rate	Recovery	Automatically logged daily from the api from google health and the inspire 3
Heart rate	Health (background)	Automatically streamed continuously from the api google health and inspire 3. Whilst considered a background non displayed metric, it will help with providing contexts to exercises etc.
Skin temperature variation	Health (background)	Automatically logged daily from the api google health and inspire 3. Whilst considered a background non displayed

		metric, it will help with confirming trends such as early sickness.
Cardio load	Health (background)	Automatically streamed daily from the api google health and inspire 3. It helps provide contexts on cardio volume to the body for trends.
Daily readiness	Health (background)	Automatically logged daily from the api google health and inspire 3. It helps provide additional context on the bodies response to the loads placed on it for further explanations of what to do.
Exercises	Health (background)	Automatically logs them from google health to help check that habits have been completed and used for other contexts. Such as checking pace and whether there are performance improvements or heart rate improvements etc.
Steps	Health (background)	Automatically logs them daily from google health to help provide contexts on cardio volume.
Active zone minutes	Health (background)	Automatically logs them daily from google health to help provide contexts on cardio volume.
Sleep duration, sleep schedule, REM sleep, Deep sleep, Sleep efficiency, time to sound sleep, sound sleep, restlessness, full awakenings, interruptions.	Health (background)	A bunch of automatically daily logged data from google health to provide more data on sleep and recovery contexts for improved advice and correlations.
VO2 max (cardio fitness score)	Performance	Automatically logs from google health when doing running exercises to provide score on cardio/lung prowess.
Running speed 5k	performance	Automatically checks from exercise running logs mentioned earlier to get their speed

		assuming the run is 5-5.5km.
Vertical jump height	performance	Manual log request from physical app. Represents explosiveness.
Plank	performance	Manual log request from physical app. Represents muscular endurance.
Deadhang time	performance	Manual log request from physical app. Represents muscular endurance as well.
Hamstring mobility	performance	Manual log request from physical app. Represents mobility.
Total energy burned	Diet (background)	Automatically logs daily from google health which provides a loose context for improved ai smart algorithms that will converge to a proper energy burn rate for use.
Food logs	Diet	Automatically checks daily from google health for logs for food, both the name and macros, to contribute to completing a energy requirement. Other micro scores are calculated in the physical app. Google health has convenient photo based logs and also conversation based minimising friction.
Profile: Age, gender, height, weight, body fat percentage	General (background) But body fat percentage is under performance	Automatically checks daily from google health for changes which provides context for improved understanding of the context of all fitness trends.
Skin health	Aesthetics	Requested from physical app, is manually completed with a third party vision model or other method to be more seamless.
Oral health	Aesthetics	Unsure how to quantify this but perhaps using a smart toothbrush rating?

Eye health	Aesthetics	Unsure how to quantify this maybe lux minutes outside?
Grooming	Aesthetics	Percentage of body hair that is considered ideally groomed, on physical app manually.
Hair density	Aesthetics	Requested from physical app to be manually completed with a third party vision model to be more seamless? Im guessing to count hairs per cm squared on scalp.
Voice quality	Aesthetics	Requested from physical app, is manually completed with a third party audio model or other method to be more seamless

I am open to more rows of data suggestions.

Ideally table 1 will exist as a scalable database bridging the platforms of physical app and health app.

Part 2: Graphs page

There are 2 ways this data will show up to the user in the physical app.

The graphs page is one of them.

Think of this as very similar to the google health's data screen. Such as vitals, sleep, fitness and likewise. The user can open the different categories and see a card table list of the data and past week, if they click on the card it opens up to a full customisable graph where the graph x axis (time) can be adjusted.

There is also a second graph option in this page at the bottom which allows the user to simply select multiple graphs for comparison and correlation. It starts off empty but users can add data types.

How will the graph y axis be standardised? Using the following next part 3:

Part 3: Body graph & Ranks page

This part is inspired by LiftRanks bodygraph/ lifting rank system.

As we are making this physical app not just to log but to provide a clear picture of one's health and to motivate them for meaningful goals of perfect physical prowess.

Arbitrary numbers on data will not suffice.

On all non -(background) data types, they will be displayed on a body graph much like those health apps that show the muscle groups.

An example is figure 1 shown below:

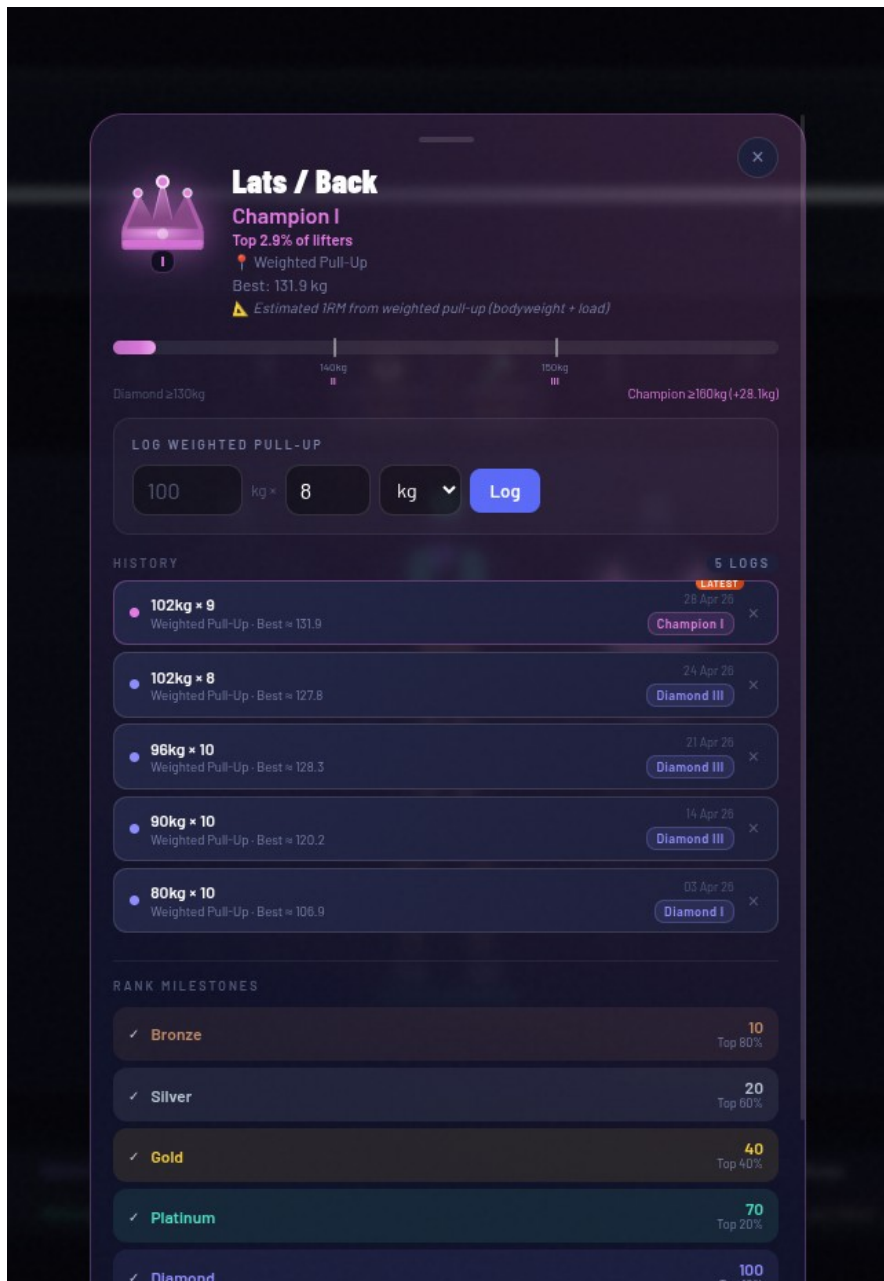
Figure 1:



The user will be able to click on these different body parts and see a pop up card on progress on their lifts.

Refer to an example image in figure 2 that elucidates this:

Figure 2:



As you can see the card shows a progress bar on the lift, a way to log it if it's a manual log and milestones.

This is also where leagues come in.

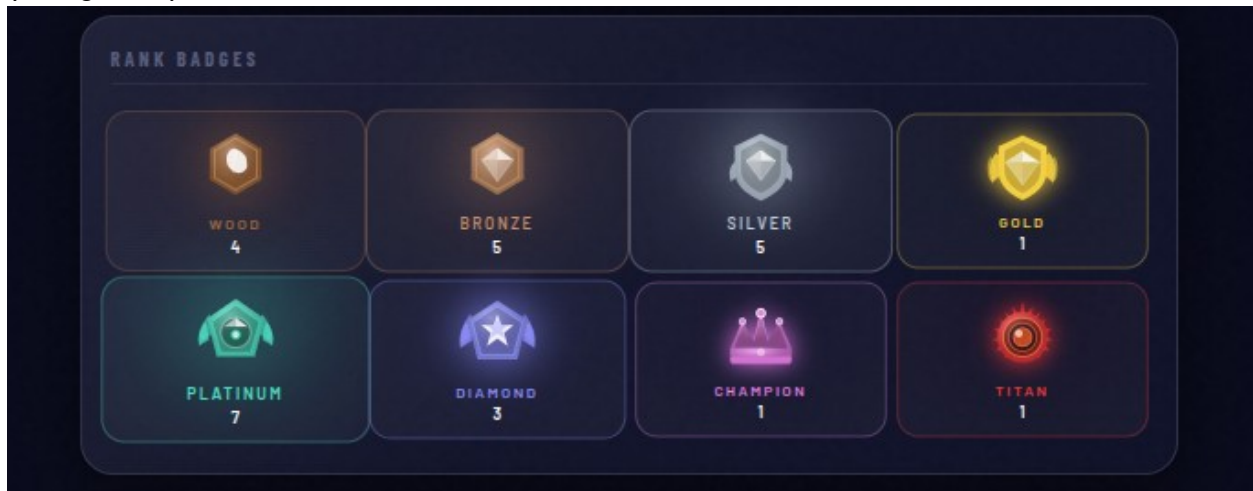
There are 8 ranks and subranks I, II and III for each metric representing thirds of progress for each. Which help show exactly where a person is in terms of fitness.

These ranks are not arbitrary.

The wood, bronze, silver, gold, platinum, diamond, champion and titan rank represent the top 99%, 80%, 60%, 40%, 20%, 10%, 3% and 1% respectively of the population.

[Uncapped Potential (The "Glory" Tier): While the Titan tier represents the top 1% of the population, a true physical pursuit requires no ceiling. Adding an uncapped fractional tier (e.g., 0.1% or Elite/Glory) provides an endless runway for a rigid fight into greatness]

This is possible because every lift or health metric is a population wide standardised measurement that can be compared for a healthy male (for now just young male in this case). For example on figure 2 you can see that latissimus dorsi rank is based on the weighted pullup/ assisted pullup total weight, with 10,20,40,70,100 kilograms for certain leagues from bronze to diamond. And any weight in between from reps is calculated to an exact 1 rep max that is then calculated to a percentile and I, II , II subrank within a rank. As a result, the user can see exactly what performance they are relative to their healthy male population to be grounded in actual meaningful goals. This applies to aesthetics, performance, recovery and strength data types that arent considered (background).



In figure 3 and 4, shown below, it can be seen how the ranks also translate into category ranks and overall ranks to also provide a holistic ranking of your physical prowess.

Figure 3:

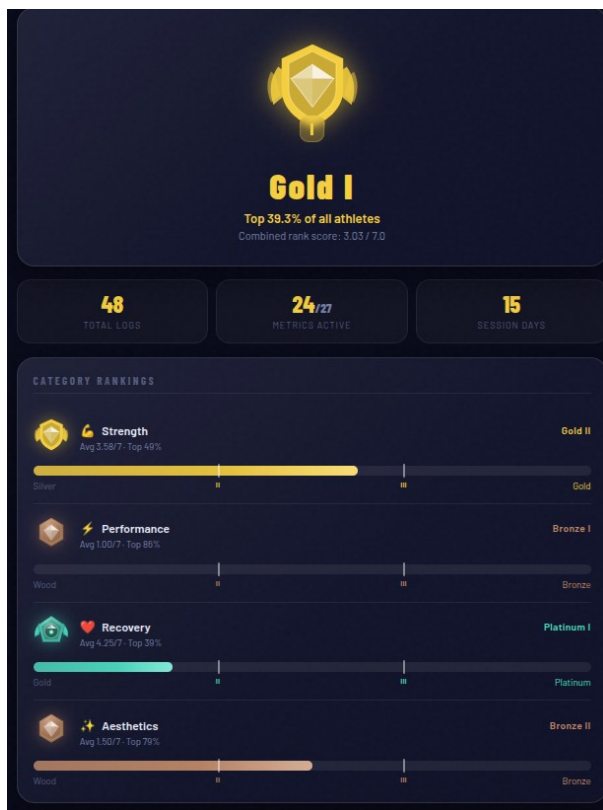
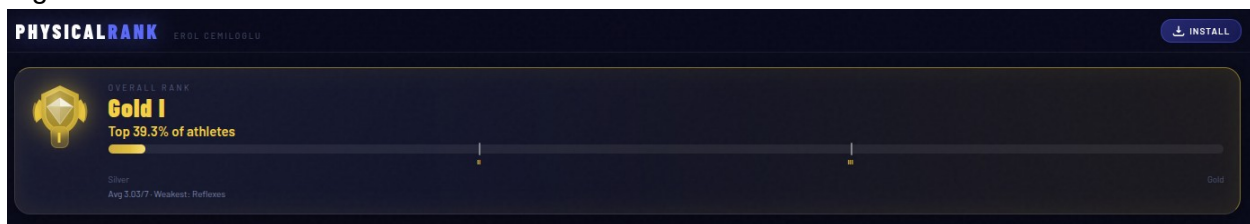


Figure 4:



You will also notice the body graph is coloured to the rank colours making a quick visual coordination of your overall relative health.

Referring back to the graphs section, figure 5 shows how the y axis will now be this ranks to standardise comparing different metrics.



Overall the user is now given percentiles and ranks that ground them in their progress and help fuel pursuable non arbitrary goals.

It will be up to us to determine the strength, recovery, performance and aesthetic standards as accurately as possible to make these ranks perfect.

It is extremely important that heavy research is done into what these standards should be for the data types because if they feel even slightly inaccurate the immersion of the gamified app is completely lost.

Part 4: Habits page

Now part 1-3 by itself is great in that it provides a robust way for a user to turn their results into scores that matter.

But a truly great physical app will find a way to continue to help the user improve and iterate to actually achieve those results and gamified scores.

We will use AI in a future part to achieve this. But first we need a system that actually records what the user is doing for the AI to keep track.

And also acts as an accountability system for the user.

My idea was to have a habits / checklist page in that the user could specify their own required goals. This would be built by them and perhaps AI when we discuss a later part, and to check them off they would have to complete manual logs or automated data from google health from actually doing the activities.

That way the AI and the user have a robust understanding of the habits/checklist set out for themselves every day (and week) but also what they did, didn't do, their volume etc.

And everyday you would get a summary of what you achieved and what you didnt on the checklist to keep you accountable. Maybe even make it more "quest"-like to gamify it.

Here is a table that helps designate how the habits/checklist sections would function. See table 2 below.

Table 2:

Habit type	How its set	How it is measured
Sleep	By default its set to reach high sleep scores which is ranked. But the user can be specific about specific concerns such as early sleep time, sleep duration, deep sleep, interruptions etc. AI could help adjust to make the goals more achievable and always iterating for improvement or suggest what to look out for. This is also connected to recovery markers data types and may propose other habits to manually check off like meditation to help reduce if allowed.	The inspire 3 automatically will measure and the google health api will provide daily logs to the physical apps regarding all this data. Then this would align against the habit goals and only check the checklist sections if it was actually achieved.
Diet	By default it is set for calorie, weight/body fat change, and protein goals automatically determined from if you wish to bulk, cut or maintain, but AI could adjust based on your meal data, changes in weight, body fat percentage and so on. I would also find it interesting if there was a diet goal that had minimum health. Like a health score for micronutrients, gut health, fibre etc. that is proportional with portion so that it can provide more robust goals for the user. The AI which will be specified more in the next part could also proscribe meal plans?	The google health api will provide the logs because the meal logs are done by google health due to the apps maturity with this software. However the physical app can be the one that further analyses micronutrients etc. for those scores.
Strength	You would be able to set a workout plan and days of the week in the habits so that it is checked off, and via logs of exercise sets it can check if you hit volume and intensity requirements and sets you need to. AI would be able to see your strength ranks progress with these habits and may propose changes to workout plan to meet goals such as if you wish to improve on your chest it can add more or less sets for	Two step verification of manual exercise rep sets places that match the habits (or if not some way to find the equivalent for it) , and also a workout exercise that is done automatically from google health which represents the same duration as the logs being placed in. may even show fatigue from the load seen from the exercise data

	chest and so on. Dynamically improving on your workout sets structure to meet your goals.	from the workout log from google health.
Performance	This one is more broad with the variety of muscle endurance as plank, mobility with hamstring stretch, cardio fitness with running 5k speed, explosiveness with jump height etc. And so the idea is if you have specific goals in mind AI will try to build a routine you need to check off here. Or of course you can set it yourself. EAch routine is connected to one of the performance types like mobility etc so it is understood which one you are trying to improve. By default AI may try to suggest improving all of the performance equally etc. Additionally simpler activities like zone minute goals, step goals can be included here too for a more holistic plan.	This would be checked off via manual checklist but also two step verification that an exercise was done during the period it was checked off +- 30 minutes to prove it was completed.
Aesthetics	May have daily/weekly requirements for measurements to be completed, and also checking off skincare, grooming, oral, ocular, vocal, and hair care routines. Each routine has a required list to write the used products/habits so that the AI knows exactly what is being used on what day so it can intelligently triangulate what could be added for improvement or how to iterate.	Manual logs would be needed for the habits but for measurements we could connect to ai models or 3rd party softwares that automatically check off the measurement habits. We could make sure the person does the habits in time though, by having a +-5 hour window to do some of the time sensitive ones or would be unable to. Not too sure if this is worth doing.
Miscellaneous	This includes activities that are often not a specific category but also useful to track if done or not. Maybe if even just for the user to benefit from like a to do list so they dont forget. Think like shaving, or journaling, or ab vacuum training, or posture training.	These are all manually checked off.

More suggestions on habit categories are appreciated.

The habits are to be in a time organised checklist with maybe a barchart explaining their location throughout the day. And the habits are set weekly to complete. I am flexible on the organisation of the presentation of this page.

Most of all, habits automatically present themselves on your google calendar, meaning it will take out of your day what you need to do minimising friction.

Part 5: The AI coach

The truly innovative, motivating and improving iteration section of this app is its AI coach. Inspired by the google health app coach, I want it to be at least as advanced and innovative as that one is.

The AI coach allows all these trends, ranked goals, habits, and background data to inform it on what is going on. Providing it with a plethora of valuable functions to drive and elucidate improvements on the user for physical perfection.

I want some structured functions for it that may be represented by contextual buttons which align it with certain data or all. Since I am the least sure of this section I am open to the most suggestions here on true perfection.

Refer to table 3 for a list of functions by this agentic AI coach.

Table 3:

AI CONTEXT	Function	How it works
Sleep	Can ask about your sleep patterns and your goals and what habits to change to improve it. Gives unprompted scheduled comments on your sleep and suggestions. May give reminders on habits you should be doing as notifications for sleep. When it requests habit changes like sleep times it should be able to agentically perform it automatically.	Would see your sleep data, and perhaps diet data especially diet times, your exercise times, heart rate, recovery markers and overall habit contexts to see what you are doing right/wrong.
Diet	Similar to sleep functions; it can align your diet with your caloric goals, or other micronutrient goals and can give suggestions, meal plans and habit changes. And change caloric requirements as it understands your weight changes to dynamically give the perfect macro	Would see your diet logs, diet times and align it to habits and overall performance and strength and weight and body fat percentage markers for overall context.

	goals.	
Exercise	Can review what exercises you have done, your exercise load, how many sets you have done, your overall volume and how it tracks with your progress. Similar initiative for reminders and changes just like sleep & diet Give changes based on load capabilities such as not being able to do enough of the checklists or not long enough it may suggest to reduce load dynamically. Or tailor workout plans completely from scratch that are placed in the habits. You can talk to the AI directly to log your sets making it even more seamless and easy.	Would see variety of exercise, cardio load and other fitness data.
Aesthetics	Can investigate the harmony and chemical perfection of your oral, haircare, or skincare routine along with your skin concerns and help procure improvements with novel research and optimum longevity. Will unprompted ask to make sure you get measurements for the data values.	Your aesthetic habit checklists and your aesthetic rank scores.
Strategic Goal setting	Can ask and tailor habit changes depending on which performance or strength etc. value you want to highlight on improving, and remember that as your emphasised goal. additionally can investigate which data values are plateauing or lagging behind and correlate habits with it or proscribe improvements. Overall it is dynamical changing its guidance based on user habits actually being completed and its ranked results.	All habits and emphasised data values.
Milestone rewarding	Can discuss progress and milestone	All data values with ranks.

and discussion	achievements, with discussions for feedback on habits and congratulations.	
Voice logging	Can use voice to talk for even more seamless connection	
General questions	Asking things like form of exercises or advice or diet choices to stay informed on the go.	
Notifications	When appropriate has notifications for habits to do and other comments.	
Habit update aware	Whenever habits are changed it will provide comments and prompt if any goal changes are desired.	
Structured categorised context.	You can see a sectioned collation of context with some of it (such as goals) deletable to be able to understand completely what the context of your AI coach has for maximum accuracy.	User perspective of context tables.
Strategic correlations	he AI should autonomously run background regressions. If it notices that a drop in deep sleep correlates with a drop in mechanical output on heavy compound lifts, it could automatically temporarily pin that graph overlay to the dashboard for viewing.	
Dynamic Volume Auto-Regulation:	If the automatically logged daily readiness score plummets, the AI should dynamically restructure the habit checklist. It could instantly modify the scheduled strength routine, lowering the volume requirement or swapping heavy lifting for the specified mobility or stretching routines.	Habits and recovery scores.

Part 6: Friends and sharing and quality of life

Ultimately, this is an app so features similar to other apps of sharing ranks, having app convenience native features and being able to connect to friends or other similar high end app features are appreciated here.

Part 7: Underlying mathematics

Below is simply inspiration, please don't copy 1 to 1 but is some of my efforts in creating how the ranking system could work on a very foundational level, you can investigate for its merits or if it can be improved:

```
// =====  
// engine/rank.js — Pure rank calculation engine  
// No DOM access. No side effects. All functions are deterministic.  
// =====  
  
import { RANKS, THRESH, MUSCLES, TOP_PCT_ARR, SUB } from '../data/metrics.js';  
  
export function epley(weight, reps) {  
  // Guard clauses for edge cases  
  if (reps <= 0 || weight <= 0) return 0;  
  if (reps === 1) return weight;  
  
  // Cap reps at 15: prevents Brzycki's divide-by-zero and keeps estimates realistic  
  const effectiveReps = Math.min(reps, 15);  
  
  // 1. Epley: Standard and aggressive  
  const epley1 = weight * (1 + effectiveReps / 30);  
  
  // 2. Brzycki: More conservative, highly accurate for low reps  
  const brzycki = weight / (1.0278 - 0.0278 * effectiveReps);  
  
  // 3. Lander: A different linear regression often used as a middle ground  
  const lander = (100 * weight) / (101.3 - 2.67123 * effectiveReps);  
  
  // Calculate the average  
  const average = (epley1 + brzycki + lander) / 3;  
  
  // Return rounded to 2 decimal places (e.g., 135.50)  
  return Math.round(average * 100) / 100;  
}  
  
// — RANK LOOKUP  
export function rankOf(mid, orm) {  
  const t = THRESH[mid] || [];  
  let r = 0;  
  t.forEach((v, i) => { if (orm >= v) r = i + 1; });  
  return Math.min(r, 7);  
}
```

```
}
```

```
export function getSubrank(mid, orm) {  
  const rank = rankOf(mid, orm);  
  const t = THRESH[mid] || [];  
  const cur = rank > 0 ? t[rank - 1] : 0;  
  const nxt = rank < 7 ? t[rank] : (t[t.length - 1] || 1);  
  if (rank === 0) { const p = (orm / (nxt || 1)) * 100; return p < 33.33 ? 0 : p < 66.66 ? 1 : 2; }  
  if (rank >= 7) return 2;  
  const p = ((orm - cur) / (nxt - cur)) * 100;  
  return p < 33.33 ? 0 : p < 66.66 ? 1 : 2;  
}
```

```
// — CONTINUOUS RANK FRACTION (0–1 within current tier)  
// Linear interpolation between the two surrounding threshold weights.  
// Used by overallScore and progress charts — ensures they are always consistent.
```

```
export function preciseFraction(mid, orm) {  
  const t = THRESH[mid] || [];  
  const rank = rankOf(mid, orm);  
  if (!orm || orm <= 0) return 0;  
  if (rank >= 7) return 1; // capped at tier top  
  const lo = rank > 0 ? t[rank - 1] : 0;  
  const hi = t[rank] ?? lo + 1;  
  return Math.max(0, Math.min(1, (orm - lo) / Math.max(hi - lo, 1)));  
}
```

```
// — PRECISE RANK VALUE — rank + continuous fraction (e.g. 3.742)  
// This is the canonical value used by both overallScore and the progress charts.
```

```
export function preciseRankValue(mid, orm) {  
  const rank = rankOf(mid, orm);  
  if (!orm || orm <= 0) return 0;  
  const t = THRESH[mid] || [];  
  if (rank >= 7) {  
    const maxT = t[t.length - 1] || 1;  
    return Math.min(7.5, 7 + (orm - maxT) / Math.max(maxT * 0.1, 1));  
  }  
  return +(rank + preciseFraction(mid, orm)).toFixed(4);  
}
```

```
export function getProgress(mid, orm) {  
  const rank = rankOf(mid, orm);  
  const t = THRESH[mid] || [];  
  const cur = rank > 0 ? t[rank - 1] : 0;  
  const nxt = rank < 7 ? t[rank] : (t[t.length - 1] || 1);
```

```

const span = Math.max(nxt - cur, 1);
const pct = rank >= 7 ? 100 : Math.max(0, Math.min(100, ((orm - cur) / span) * 100));
return {
  orm, rank, cur, nxt, pct,
  m1: +(cur + span * 0.333).toFixed(1),
  m2: +(cur + span * 0.666).toFixed(1),
};
}

```

```

export function overallScore(logs) {
  const scores = MUSCLES.map(m => preciseRankValue(m.id, latestORM(m.id, logs)));
  return scores.reduce((a, b) => a + b, 0) / scores.length;
}

```

```

// — LATEST ORM HELPER
export function latestORM(mid, logs) {
  const e = logs[mid];
  return (e && e.length) ? e[e.length - 1].oneRM : 0;
}

```

```

// — PRECISE PERCENTILE INTERPOLATION
export function getPrecisePercent(mid, orm) {
  if (!orm || orm <= 0) return '99.9';
  const rank = rankOf(mid, orm);
  const t = THRESH[mid] || [];
  if (rank >= 7) {
    const maxThresh = t[t.length - 1];
    const over = orm - maxThresh;
    return Math.max(0.01, 1 - (over * 0.05)).toFixed(2);
  }
  const curW = rank > 0 ? t[rank - 1] : 0;
  const nxtW = t[rank] || 1;
  const curP = TOP_PCT_ARR[rank];
  const nxtP = TOP_PCT_ARR[rank + 1];
  const fraction = (orm - curW) / (nxtW - curW);
  return (curP - (fraction * (curP - nxtP))).toFixed(1);
}

```

```

export function getPrecisePercentOverall(score) {
  const baseRank = Math.floor(score);
  if (baseRank >= 7) return '1.0';
  const fraction = score % 1;
  const curP = TOP_PCT_ARR[baseRank];
  const nxtP = TOP_PCT_ARR[baseRank + 1];
}

```

```
    return (curP - (fraction * (curP - nxtP))).toFixed(1);
}
```

```
// — DATE PARSING (en-AU format: "02 Apr 25")
```

```
export function parseDate(str) {
  const months = { Jan:0, Feb:1, Mar:2, Apr:3, May:4, Jun:5, Jul:6, Aug:7, Sep:8, Oct:9, Nov:10,
Dec:11 };
  const p = str.split(' ');
  if (p.length < 3) return new Date();
  return new Date(2000 + parseInt(p[2]), months[p[1]] || 0, parseInt(p[0]));
}
```

```
// — DISPLAY FORMATTING
```

```
export function formatScore(mid, val, unit) {
  if (!val && val !== 0) return '—';
  const m = MUSCLES.find(x => x.id === mid);
  if (m && m.type === 'score') return val.toFixed(1) + ' ' + m.unit;
  return val.toFixed(1) + ' ' + (unit || 'kg');
}
```

```
// — SUB-RANK STRING HELPER
```

```
export function subLabel(mid, orm) {
  return SUB[getSubrank(mid, orm)];
}
```

```
// — RANK COLOR HELPERS (reads from RANKS, no DOM)
```

```
export function rankColor(mid, logs) {
  return RANKS[rankOf(mid, latestORM(mid, logs))].c;
}
```

```
export function rankGlow(mid, logs) {
  return RANKS[rankOf(mid, latestORM(mid, logs))].glow;
}
```